DESIGN PITCH

AI-Assisted Fast-Track Screening for Initial SSI/SSDI Disability Claims

Record-based screening to widen the fast track, with human review of every denial

Team 06 · Frontier

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Illustrative cases

Joe and Peter, two fictional applicants

Limited Fast-Track Coverage of Initial Disability Claims

—— *Evidence from SSA administrative data, with an illustrative case*

209 days
average wait for a first decision

37%
of claims approved at the first decision

8.3%
fast-tracked by QDD + CAL (FY2023)

Diagnosis

- Fast tracks read only the application form, before records arrive.
- So most clear approvals still get a full review from scratch.

J Illustrative case: Joe

1 Leaves the police force after a spinal cord injury.

2 Form reports "back injury, uses a wheelchair."

3 Neither fast track flags him: they only see the form.

4 Waits for an examiner to review his medical records.



Research question. Can record-based AI screening expand fast-track coverage of initial SSI/SSDI claims without lowering accuracy against an independent expert standard?

Sources: SSA Commissioner's letter (Nov 24, 2025); SSA Fast-Track Public Use File; SSDI Annual Statistical Report 2024 (Table 61); SSA DDS Net Accuracy.

Mechanism: AI Fast-Tracks Clear Approvals

—— *Mechanism: decision-support automation; a human decides every denial*

Theory of change.
AI applies SSA criteria to medical records.
It approves clear matches; examiners review the rest and decide all denials.

Causal chain

Input	Activity	Output	Short-term	Outcome:
Application + medical records	AI checklist, SSA rules, score	Clear → fast track; Rests → human	Fewer full reviews by human	Faster decisions; no AI denials

J Illustrative case: Joe at each step

Joe's form + 40-page medical file	Spinal cord injury,; No earned income	Meets SSA criteria → fast track	Examiner checks 3 quotes, signs in minutes	Joe decided sooner; time freed for Peter's file

Supporting evidence: about half of SSA approvals meet or equal a medical listing

- QDD: 93% of its fast-tracked sample was allowed
- SSA already reads records with AI (IMAGEN), but has published no evaluation

Sources: SSDI Annual Statistical Report 2024 (Table 64); SSA OIG A-01-09-19030; ACT-IAC (IMAGEN). Listing 11.08 = spinal cord disorders.

Record-Based Fast-Track Screening for Initial Claims

Closest existing products (QDD and Compassionate Allowances), screen only the application form

1 Collect → **2 Read: AI checklist** → **3 Score: SSA rules** → **4 Route** → **5 Decide**

J Case A · Joe → fast track

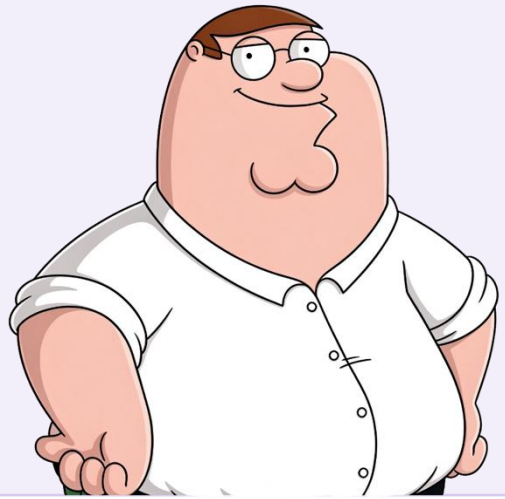
Field	AI copies from the file	Page
Diagnosis	complete spinal cord injury	p. 12
Function	cannot stand or walk	p. 15
Work	no work since 2025	p. 2
Rule: meets Listing 11.08 → fast track		

Examiner: checks the 3 quotes, approves

P Case B · Peter → human review

Field	AI copies from the file	Page
Diagnosis	lower-back strain	p. 3
Treatment	two ER visits, no specialist	p. 5–6
For approval	"cannot lift over 10 lb"	p. 6
Rule: no listing met; can still work → human review		

Examiner: decides; the AI never denies



Current practice: QDD + Compassionate Allowances

- Read the application form: 8.3% fast-tracked

Proposed tool

- Reads the medical records: target ≥ 15% fast-tracked

Sources: POMS DI 23022.010; SSA OIG A-15-11-11175; SSA Fast-Track Public Use File (FY2023). Page numbers refer to our made-up case files.

Specification: User, Task, and a Measurable Claim

—— *Planned targets and evaluation design*

User

State disability examiner
--making first decisions on adult claims

Task

- Fast-track clear approvals.
- Send other cases to human review
- with AI-highlighted evidence.

Targets and requirement

- Aim: ~15% fast-tracked
- Aim: ≥95% confirmed eligible
- All denials decided by a human

Evaluation design

1 Write

50 fictional cases covering clear, uncertain and ineligible claims.

2 Calibrate

Use 10 cases to adjust the tool, then lock settings and register the test plan on OSF.

3 Test

Test 40 new cases. We plan to use experienced examiners' decisions as our reference.

Expected routing in example cases

Joe → fast track, with human confirmation
Peter → full human review, with AI evidence
Joe, still working → full human review

What we will measure

Coverage: fast-track cases / all test cases
Accuracy: eligible cases / fast-track cases
Targets to test, not proven results

Risk Anticipation: Two Primary Risks and Design Responses

—— What could go wrong, how we would notice, and what we do about it

P R1 Wrong approvals on the fast track

WHAT GOES WRONG

Peter's ER note is read as a permanent limit, so he slips onto the fast track.

WARNING SIGN

Pilot accuracy below 95%

OUR RESPONSE

Raise the threshold: fast-track fewer claims

J R2 The AI misreads or invents evidence

WHAT GOES WRONG

The AI 'finds' a spinal cord injury that isn't in Joe's file.

WARNING SIGN

A quote doesn't match the record

OUR RESPONSE

Drop the item, and **send the case to a human**

R3 Our answer key could be wrong

Teammates **re-check it blind**; disputed cases are rewritten.

R4 Thin records never qualify

We report fast-track rates by impairment; the slow lane is **never a denial**.

Our commitment: the AI never denies anyone. A human examiner decides every denial.

Budget: $200 Course Plan and a Scaling Budget for SSA

This project: our team's AI use ($200)

Item	Model, $/M tokens	$ / call	Calls	Total
Claude Pro: building, writing	$20–23 / mo	fixed	2 mo	**$46**
API: running the tool	Sonnet 5.5, $2 / $10	0.04	150	**$6**
API: building the tool	Sonnet 5.5, $2 / $10	0.021	1,400	**$29**
Testing: pilot, demos, stress	Sonnet 5.5, $2 / $10	0.021	2,000	**$42**
Testing: Opus double-check	Opus 5.5, $4 / $20	0.042	300	**$13**
Testing: Fable audit	Fable 5.1, $10 / $50	0.14	250	**$35**
Reserve				**$29**
Total				**$200**

Largest line: Claude Pro, $46
Per call: 2¢ (Sonnet) to 14¢ (Fable)
Reserved for testing: $90, 45% of the cap

If SSA adopted it: scaling budget (estimates)

Step	Scope	Cost
1 Follow-on study	9 months, real files, 3 experts	**≈ $43k**
2 State pilot	1–2 state DDS offices, 1 year	**≈ $1–1.5M**
3 National rollout	≈ 2.36 M initial claims a year	**≈ $3M / yr**
Benefit	Examiner time freed ≈ 84,000 more claims a year	**≈ $25M / yr**

National cost per year ≈ $3M (range $2–6M)

- **AI:** ≈ $1.2M (≈ 50¢ per 200-page file)
- **Audits of fast-tracked claims:** ≈ $1M
- **Yearly independent expert check:** ≈ $0.15M
- **Secure hosting, maintenance:** ≈ $0.75M

One-time build ≈ $1M, similar to SSA's AI upgrade of IMAGEN

Our ask: fund a real-file study (≈ $43k) and a one-year state pilot for our AI-assisted screening tool.
—— More people like Joe get faster answers, and every denial is still decided by a person.

Course: Anthropic list prices (Oct 22, 2026); Claude Pro $20/mo + tax. Scaling: SSA FY2025 plan (claims, 276 cases per work-year); TMF IMAGEN investment ($0.99M); assumed 4 h saved per fast-tracked claim at $40/h. All scaling figures are rough estimates.